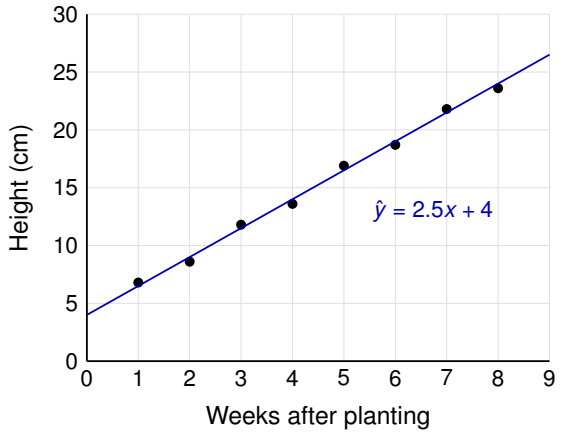
Slope, Intercept, and Correlation KEY

Name _____

A student measured a plant every week. The line of best fit is $\hat{y} = 2.5x + 4$.



Slope (2.5): For each additional week, the predicted height increases by **2.5 cm**.
y-intercept (4): When $x = 0$ (the day it was planted), the predicted height is **4 cm**.

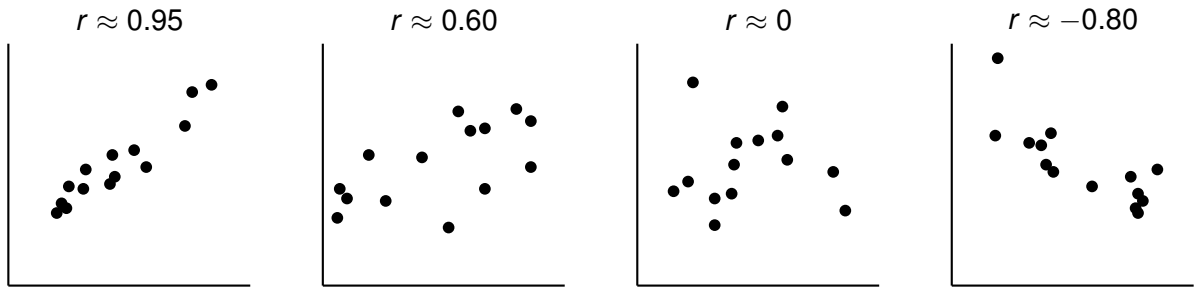
Sentence frames

Slope: For each additional [x unit], the predicted [y] increases (or decreases) by [slope and units].

Intercept: When [x] is 0, the predicted [y] is [intercept and units].

Gotcha: Sometimes the intercept has no real meaning. If x is a person's height, $x = 0$ is not possible.

The correlation coefficient r measures how strong a **linear** pattern is.



r is always from -1 to 1.

Sign shows the direction: $r > 0$ positive, $r < 0$ negative.

Size shows the strength: close to 1 or -1 is strong. Close to 0 is weak.

$r = -0.8$ is **stronger** than $r = 0.6$. The sign only tells the direction.

Correlation is not causation. In summer, ice cream sales and sunburns both go up. They have a strong positive correlation. Ice cream does not cause sunburns. Hot, sunny weather causes **both**. A hidden cause like this is called a **lurking variable**.

Work with a partner. Compare answers when you finish each problem.

1. The value of a used car is $\hat{y} = -2.5x + 30$, where x is the age in years and y is the value in thousands of dollars.

   (a) Explain the slope in context.
   For each additional year of age, the predicted value decreases by $2,500 (2.5 thousand dollars).

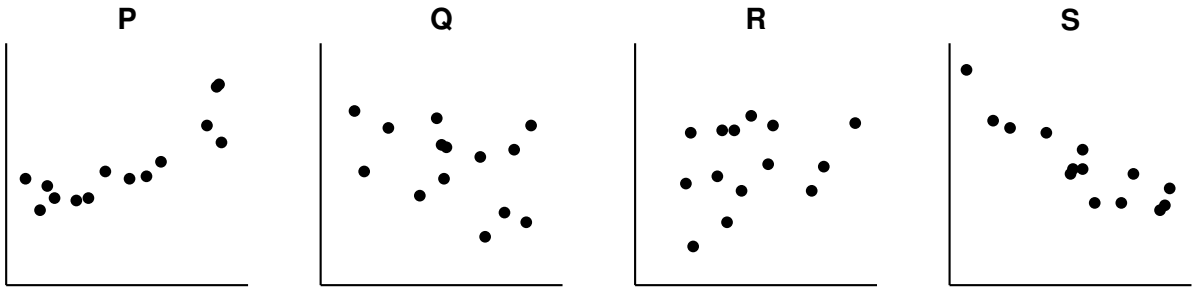
   (b) Explain the y-intercept in context.
   When the car is new (0 years old), the predicted value is $30,000.

2. For adults, arm span can be predicted from height: $\hat{y} = 1.02x - 1.3$, where x and y are in inches.

   (a) Explain the slope in context.
   For each additional inch of height, the predicted arm span increases by 1.02 inches.

   (b) Does the y-intercept have a real meaning here? Explain.
   No. It says a person 0 inches tall has an arm span of -1.3 inches. No adult is 0 inches tall.

3. Match each r to a scatterplot: $r = -0.9$, $r = 0.3$, $r = 0.85$, $r = -0.4$



P: $r = 0.85$. Q: $r = -0.4$. R: $r = 0.3$. S: $r = -0.9$.

4. Is each statement true or false? Explain.

   (a) $r = -0.9$ shows a weaker relationship than $r = 0.5$.
   False. -0.9 is closer to -1, so it is stronger. The minus sign only means negative.

   (b) A scatterplot has $r = 1.3$.
   False. r is always from -1 to 1.

   (c) $r = 0$ means there is no linear pattern.
   True. (There could still be a curved pattern.)

5. In a city, the number of fire trucks sent to a fire and the amount of damage have a strong positive correlation.

   (a) Do more fire trucks cause more damage? Explain.
   No. Correlation does not prove cause.

   (b) What lurking variable could cause both?
   The size of the fire. A big fire causes more damage and gets more trucks.

6. A study finds that students who eat breakfast have higher grades. A news story says, "Eating breakfast raises your grades." Is the news story right? Name a lurking variable that could explain the pattern.

   Not proven. This only shows an association. Lurking variables: more sleep, a family routine, or more support at home could lead to both breakfast and better grades.